
Algorithm 3.1: PPO high-frequency discrete trading loop with explicit vector state

```
1 procedure TRAIN-TRADING-AGENT( $\{P_t\}, c, E, K, C_0$ )
2    $\triangleright \{P_t\}$ : price series divided into  $K$  daily episodes
3    $\triangleright c$ : proportional transaction-fee coefficient
4    $\triangleright E$ : training epochs;  $C_0$ : initial cash
5   Initialise policy parameters  $\theta$  and value parameters  $\phi$  randomly
6   for  $epoch\ e \in \{0, \dots, E-1\}$  do
7     Initialise trajectory buffer  $\mathcal{B} \leftarrow \emptyset$ 
8     for  $day\ k \in \{1, \dots, K\}$  do
9        $\mathcal{B} \leftarrow \mathcal{B} \cup \text{ROLLOUT-EPISODE}(\{P_t^{(k)}\}, \theta, c, C_0)$ 
10      Update  $\theta$  by maximising the clipped PPO objective  $L^{\text{CLIP}}(\theta; \mathcal{B})$ 
11      Update  $\phi$  by minimising the value loss  $L^V(\phi; \mathcal{B})$ 
12    return  $\theta^*, \phi^*$ 

13 procedure ROLLOUT-EPISODE( $\{P_t\}, \theta, c, C_0$ )
14   Initialise cash  $C_0$ , shares  $n_0 \leftarrow 0$ , and  $s_0 \leftarrow [\Delta P_0, C_0, n_0]^\top$ 
15   Initialise daily buffer  $\mathcal{B}_{\text{day}} \leftarrow \emptyset$ 
16   for  $minute\ t \in \{0, \dots, T-1\}$  do
17      $\triangleright T = 390$  trading minutes
18     Compute valid-action mask  $m_t$  from cash  $C_t$  and shares  $n_t$ 
19      $\triangleright$  Hold is always valid; Buy needs cash; Sell needs one share
20     Compute masked action probabilities  $\tilde{\pi}_\theta(a_t | s_t, m_t)$ 
21     Sample  $a_t \sim \tilde{\pi}_\theta(a_t | s_t, m_t)$ , where  $a_t \in \{0:\text{Hold}, 1:\text{Buy}, 2:\text{Sell}\}$ 
22     Execute one-share transition: 
$$\begin{cases} C_{t+1} = C_t, & n_{t+1} = n_t, & a_t = 0, \\ C_{t+1} = C_t - P_t(1+c), & n_{t+1} = n_t + 1, & a_t = 1, \\ C_{t+1} = C_t + P_t(1-c), & n_{t+1} = n_t - 1, & a_t = 2; \end{cases}$$

23     Observe  $\Delta P_{t+1} = (P_{t+1} - P_t)/P_t$ 
24     Compute portfolio wealth  $W_t = C_t + n_t P_t$  and reward  $r_t = W_{t+1} - W_t$ 
25     Construct  $s_{t+1} \leftarrow [\Delta P_{t+1}, C_{t+1}, n_{t+1}]^\top$ 
26     Store  $(s_t, a_t, r_t, s_{t+1}, m_t)$  in  $\mathcal{B}_{\text{day}}$ 
27   return  $\mathcal{B}_{\text{day}}$ 
```

Interpretation. C_t is cash, n_t is the number of shares held, and $W_t = C_t + n_t P_t$ is portfolio wealth. The action mask prevents infeasible buys and sells. Transaction fees enter the cash transition, so they are automatically reflected in the wealth-change reward.